AI-BASED EVERYDAY HEALTH COMPANION

Personalized Health Monitoring & Insight Generation

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Problem Statement & Motivation

Existing healthcare tracking systems mainly record isolated health information without providing personalized behavioral analysis. Most platforms lack AI-driven insights capable of identifying relationships between mood, stress, and sleep behavior over time, making wellness patterns difficult to interpret from raw health data alone.

Mental health, stress, and sleep significantly impact overall well-being, yet many existing wellness applications fail to provide meaningful behavioral interpretation. This project combines AI-assisted analytics, NLP sentiment analysis, and interactive visualization to create a personalized wellness monitoring platform.

System Architecture

- **User** — Users enter daily wellness information including mood, stress, sleep patterns, symptoms, medication intake, and journal reflections through the application interface.
- **Streamlit Frontend** — The **Streamlit frontend** provides an interactive and user-friendly dashboard for account management, daily health logging, visualization, and personalized wellness monitoring.
- **FastAPI Backend** — The **FastAPI backend** handles API communication, authentication, data processing, analytics requests, and communication between the frontend and database.
- **AI Analytics Engine** — The **AI analytics engine** performs rule-based analysis, NLP sentiment analysis, trend detection, correlation analysis, and wellness risk scoring to generate personalized health insights.
- **PostgreSQL Database** — The **PostgreSQL database** hosted through Supabase securely stores user profiles, wellness logs, mood records, sleep data, symptom tracking, and journal entries.
- **Results Dashboard** — The **results dashboard** presents AI-generated wellness summaries, behavioral trend graphs, sentiment analysis results, and interactive health visualizations for users.

Related Work

NLP & Sentiment Analysis in Mental Health

- Studies showed that NLP and sentiment analysis can identify emotional and behavioral patterns from journals and textual health data.

Explainable AI in Healthcare Systems

- Research emphasized the importance of transparent and interpretable AI systems in improving healthcare understanding and user trust.

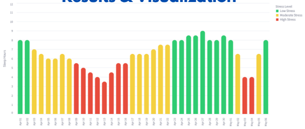
Behavioral Trend Analytics

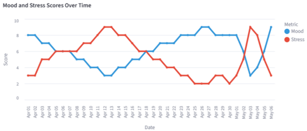
- Recent AI-driven healthcare systems demonstrated the effectiveness of combining behavioral analytics and personalized wellness insight generation.

AI-Based Wellness Monitoring Systems

- Recent AI-driven healthcare systems demonstrated the effectiveness of combining behavioral analytics and personalized wellness insight generation.

Results & Visualization





Interactive visual analytics demonstrated behavioral relationships between mood, stress, and sleep patterns while improving wellness interpretation

Key Findings

- The system successfully identified **emotional and behavioral wellness patterns** using AI-driven trend monitoring.
- Higher stress levels frequently aligned with declining mood and reduced sleep behavior.
- Weekly comparison and interactive graphs improved **wellness interpretation and behavioral trend detection.**
- Journal reflections were analyzed to identify **positive, neutral, and negative emotional tone.**

Future Work & Conclusion

Future enhancements of the system include integrating machine learning-based predictive analytics, advanced NLP models, wearable device connectivity, mobile application deployment, and personalized recommendation systems to improve behavioral wellness monitoring and healthcare analysis.

The AI-Based Everyday Health Companion demonstrates how AI-driven behavioral analytics and NLP techniques can improve wellness monitoring through personalized insights, trend analysis, and interactive health visualization. By combining mood, stress, sleep, and journal analysis, the system provides a more understandable and user-centered approach to digital wellness tracking.

References

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